Methodology

1. **Data Alignment & Cleaning:** Ingested Historical Trader Data and Fear/Greed Index. Converted Hyperliquid `Timestamp IST` to standard dates and merged the datasets on a daily level. Both datasets were verified to have zero missing values and zero duplicate rows.
2. **Feature Engineering:** Aggregated execution-level data into daily metrics per account. Engineered features include Daily PnL, Win Rate, Trade Frequency, Average Trade Size (used as a proxy for leverage/risk exposure), and Long/Short Ratio.
3. **Segmentation & Machine Learning:** Grouped accounts into basic behavioral segments using median thresholds. Furthermore, developed a **K-Means Clustering** algorithm to automatically categorize users into three advanced archetypes (Scalpers, Whales, Casual). Finally, trained a **Random Forest Classifier** to predict next-day profitability.

Key Insights

1. **Performance Asymmetry:** While win rates remain remarkably stable across market sentiments (~61%), the *Average Daily PnL* is significantly higher during 'Fear' days ($5,185) compared to 'Greed' days ($4,144). 'Fear' markets are driven by massive blow-out days for specific top traders.
2. **Behavioral Shifts:** Traders adapt aggressively to sentiment. During 'Fear' days, average trade frequency jumps to 105 trades per account (vs. 77 in Greed), and average trade size spikes by ~43%.
3. **Archetype Divergence:** High-Win-Rate traders thrive in 'Fear' markets, drastically increasing their trade size and capturing volatility. Conversely, Low-Win-Rate traders struggle during 'Fear' markets but perform much better in trending 'Greed' environments.

Strategy Recommendations

Based on the data, here are two proposed platform rules of thumb:

1. Contextual Risk Constraint (The "Fear" Rule)

- *Rule:* During 'Fear' market days, algorithmically enforce a reduction in maximum allowable position size limits for the *Low-Win-Rate / Infrequent* trader segment. Conversely, dynamically increase position limits for *High-Win-Rate* accounts.
- *Why:* Protects inconsistent traders from liquidation during high-volatility events, while maximizing volume and fees from traders who demonstrably capitalize on Fear swings.

2. Regime-Based Liquidity Incentives (The "Greed" Rule)

- *Rule:* When the sentiment index shifts to 'Greed', overall trading frequency naturally drops. The platform should deploy targeted UI alerts or temporarily increase maker-rebates specifically for the *Frequent* trader segment.
- *Why:* Incentivizes active users to provide limit orders, ensuring the order book remains sufficiently deep even when the broader market slows down.